

HAND GESTURE RECOGNITION USING MEDIAPIPE & MACHINE LEARNING

Team Members:

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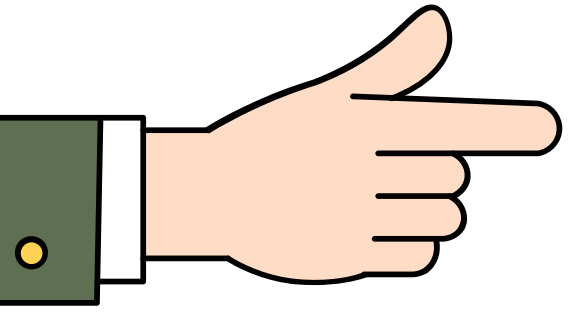




Introduction

- ➔ A system that detects hand gestures in real-time
- ➔ Uses MediaPipe for landmark extraction
- ➔ Custom dataset for training
- ➔ ML model for classification
- ➔ Flask web app for displaying live prediction





Problem Statement

- Need for Accuracy
- Requires stable landmark detection
- Hand gestures vary by user
- Must work in different lighting conditions

- User Accessibility
- Should be easy to use
- Runs in browser, no extra installation
- Helpful for gesture-based communication systems



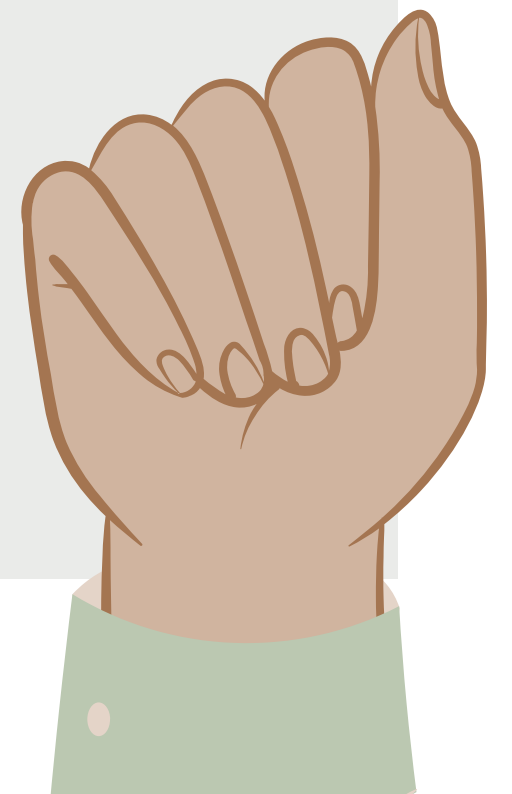
Datasets and Tools Used

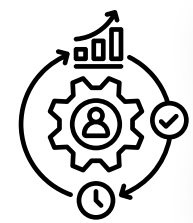
➤ Dataset

- Custom dataset
- 100 images per alphabet gesture
- Contains hand landmark coordinates
- Preprocessed into numerical features

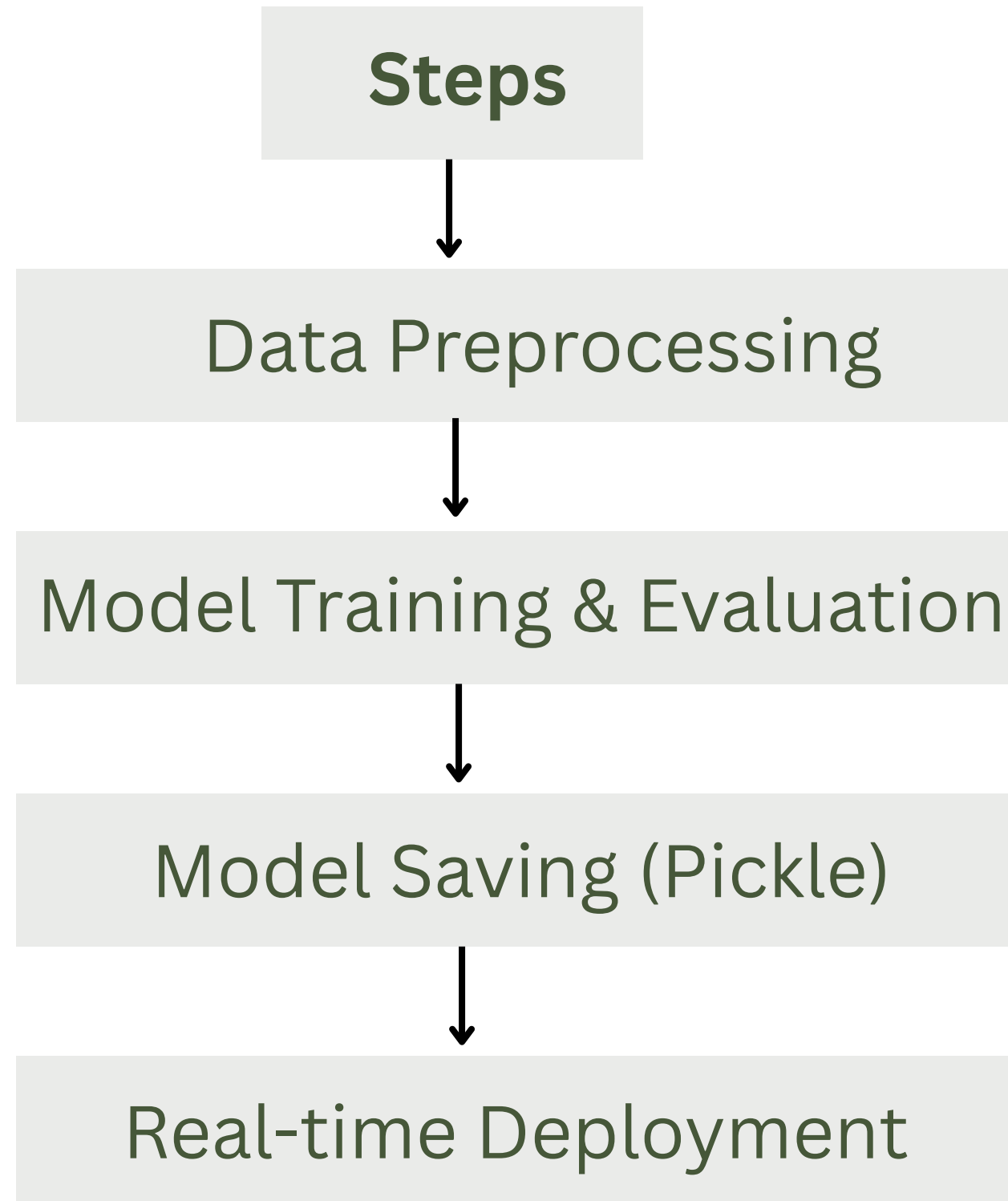
➤ Tools

- MediaPipe
- OpenCV, NumPy
- Scikit-learn
- Flask
- gTTS (Google Text-to-Speech)





Data Performance Pipeline





Methodology - System Workflow

1. User clicks
'Start Camera'

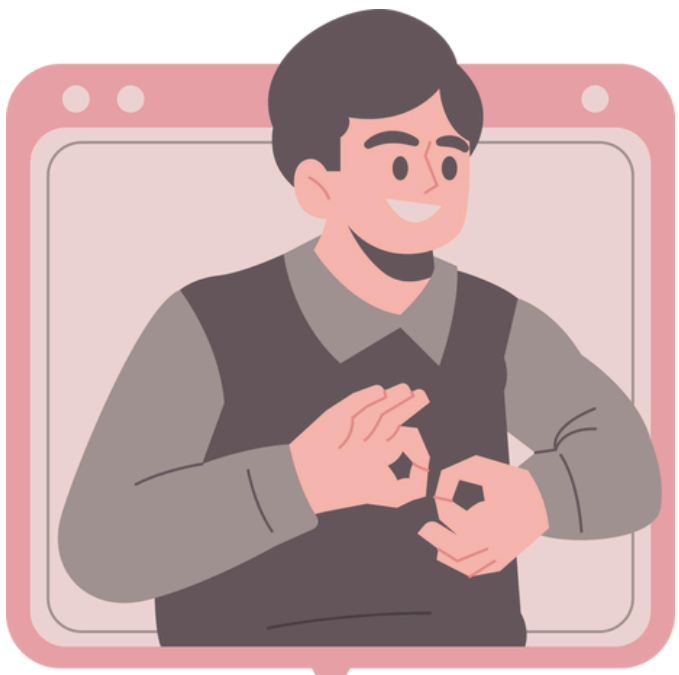
2. Webcam opens
through Flask

3. MediaPipe detects
hand

4. Extracts 42
landmark features

6. Result displayed
on video stream

5. ML model
predicts gesture





Data Preprocessing

1. Extract 21 landmarks $\rightarrow$ (x, y) $\rightarrow$ total 42 features

2. Normalize values using minimum coordinate subtraction

3. Store feature list in CSV

4. Use for training model





Algorithms Used

MediaPipe Hands for
landmark detection

OpenCV for video
processing

RandomForest Classifier

Pickle for saving model

Flask for deployment

gTTS — Google text-to-
Speech Output



Evaluation Metrics

Training Accuracy: 100.00%

Testing Accuracy: 99.62%

Confusion metrics

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Result and Observations

Model predicts gestures accurately

MediaPipe gives stable landmark tracking

Performs well in real-time

Works in browser without installation

Sound output gives audio feedback for each gesture

Slight errors in poor lighting conditions



Frontend and Backend Implementation

Frontend

Simple HTML interface

Buttons: Start, Stop, Exit

Live video feed displayed

Gesture text shown on screen

AJAX functions to fetch predictions

Backend

Flask routes

/start_camera

/stop_camera

/video_feed

/get_gesture

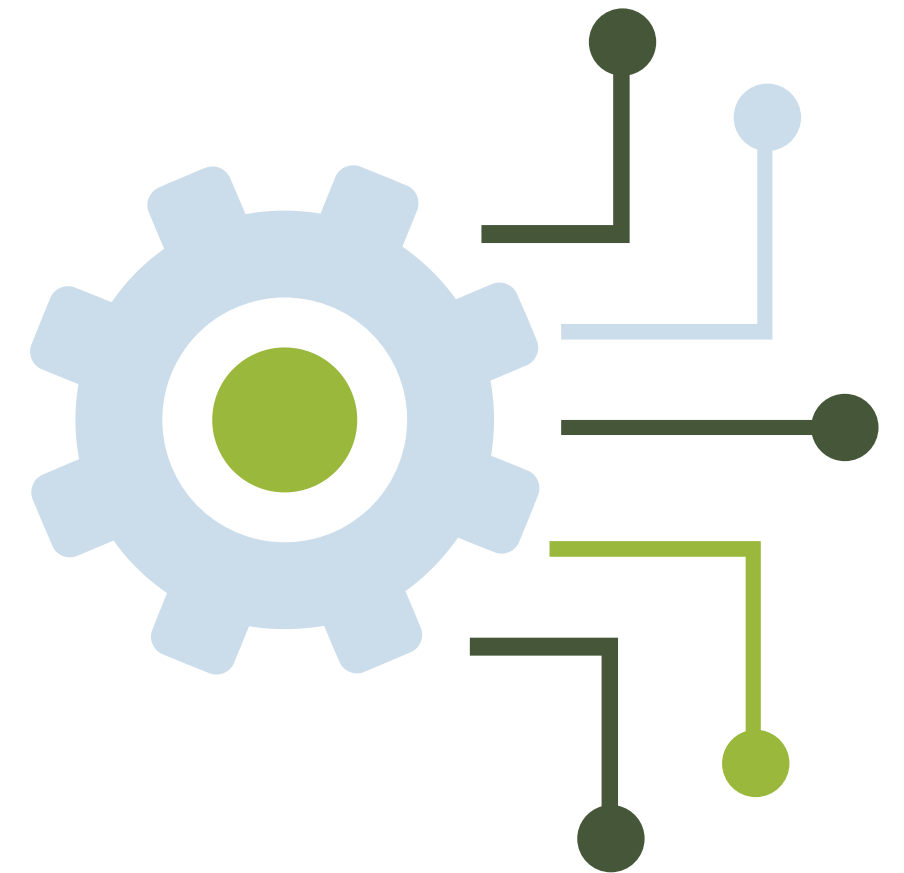
Real-time frame generation using generator function

Predicts gesture using trained model



System Implementation

- **Frontend (Browser)** -> HTML, CSS, JavaScript
Displays camera stream
- **Backend (Flask App)** -> Runs MediaPipe
Predicts gesture
Sends video to frontend
- **Machine Learning Model** -> Custom dataset
Pickle model loaded in backend



Conclusion



The system successfully recognizes hand gestures with good accuracy. It works fast, is lightweight, and easy to access. With the voice output feature, the system becomes even more interactive and user-friendly. In future, this project can be extended for IoT control, full sign-language translation, mobile deployment, and many more advanced real-time applications.

Future Scope

- ✓ Add voice feedback for words/sentences
- ✓ Convert multiple gestures → speech
- ✓ Deep learning for higher accuracy
- ✓ Two-hand gesture recognition
- ✓ Mobile app version
- ✓ Smart-home control (gesture + voice)
- ✓ Full sign language translator



THANK YOU

